

AI & Prompt Engineering

Assignment_2

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Track: Web & UI

Three Advanced Techniques for Improving Generative AI Responses

(Not Covered in the Course)

1. Answer-Then-Critique (Self-Review Prompting)

Answer-Then-Critique is a technique where the model is explicitly instructed to:

1. Produce an initial response
2. Critically review its own output
3. Identify weaknesses, gaps, or risks
4. Provide an improved version

This turns the model into both a **generator** and an **evaluator**.

Example Prompt

“Provide an answer to the problem, then critically evaluate your response for clarity, completeness, and potential errors. Finally, provide a revised improved version.”

Why It Improves AI Responses

- Forces **reflection instead of first-pass output**
- Reduces hallucinations and shallow explanations
- Encourages structured reasoning and refinement
- Mimics real-world expert workflows (draft → review → revise)

Why It's Distinct

Unlike chain-of-thought, this technique focuses on **quality assurance**, not reasoning transparency.

2. Constraint-Saturation Prompting

Constraint-Saturation involves deliberately applying **multiple, sometimes competing constraints** to guide the model toward a precise and realistic output.

Constraints may include:

- Format limitations
- Tone restrictions
- Domain boundaries
- Length caps
- Forbidden assumptions

Example Prompt

“Explain JWT authentication in React in under 150 words, using no code, avoiding backend terminology, and targeting a non-technical audience.”

Why It Improves AI Responses

- Prevents over-generalization
- Forces prioritization of key ideas
- Produces **focused and audience-specific** outputs
- Improves usability in real-world scenarios

Why It's Distinct

Most courses teach narrowing **topics**, not **output behavior under dense constraints**

3. Perspective Switching (Multi-Role Reasoning)

Perspective Switching instructs the model to analyze the same problem from **multiple professional or stakeholder viewpoints**, then synthesize a final response.

Example Prompt

“Analyze this authentication design from the perspectives of:

1. A security engineer
2. A frontend developer
3. A product manager

Then provide a final balanced recommendation.”

Why It Improves AI Responses

- Reduces single-angle bias
- Surfaces hidden trade-offs
- Produces more realistic and implementable solutions
- Enhances decision-making quality

Why It's Distinct

Unlike role prompting, this technique requires **simultaneous reasoning across roles**, not a single assumed persona.

Conclusion

These three techniques — **Answer-Then-Critique**, **Constraint-Saturation**, and **Perspective Switching** — go beyond standard prompt engineering methods by introducing **self-evaluation**, **behavioral control**, and **multi-stakeholder reasoning**.

They significantly improve response accuracy, depth, and real-world relevance, especially in professional and high-stakes applications.